GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Chemical Engineering

Subject Code: DI05005011

Subject Name: Mass Transfer - II

w. e. f. Academic Year:	2026-27
Semester:	V
Category of the Course:	PCC

Prerequisite:	Student should have knowledge regarding Unit Operations and Basic Chemistry
Rationale:	Mass transfer is crucial in chemical industries as it corresponds to many separation, purification, and reaction processes. Diploma Chemical Engineers play a crucial role inoperating equipment and machinery, monitoring process parameters, ensuring smooth production flow, supervising specific sections of a chemical plant including mass transfer operations, ensuring process control, quality control, safety and environment regulations.Using the fundamental concepts mass transfer the basic mass balance equations of mass transfer operations and separation of molecules can be estimated. The knowledge of the basic laws and principles of mass transfer helps to operate smooth and proper operation of various mass transfer operations like Distillation, Drying, Humidification, Crystallization, Adsorption and Ion Exchange. Hence the course has been designed to develop this competency and its associated cognitive, practical and affective domain learning outcomes.

Course Outcome:

After successful completion of this course, students will be able to:

No	Course Outcomes	RBT Level
01	Apply fundamental concepts of mass transfer operations in engineering problems.	R, U, A
02	Analyze the working principles and applications of various mass transfer equipment.	R, U, A
03	Compute material balance for mass transfer processes under different operating conditions.	A

**Revised Bloom's Taxonomy (RBT)*

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Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Equipment for Gas Liquid Operations	1.1 Importance of Gas-Liquid Operations 1.2 Classification of Equipment for Gas-Liquid Operations 1.2.1 Gas Dispersed (Construction &Working) 1.2.1.1 Mechanically Agitated Vessel 1.2.1.2 Tray Tower 1.2.2 Liquid Dispersed (Construction &Working) 1.2.2.1 Spray Tower 1.2.2.2 Packed Tower 1.3 Types of Tray and Packing 1.4 Operating Problems of Tray Tower and Packed Tower 1.5 Real Tray & Tray efficiency- Point Efficiency, Murphy Efficiency, Overall-Tray efficiency	04	09
Unit – II Distillation	2.1 Distillation and Its Industrial Applications 2.2 Vapor Liquid Equilibria 2.2.1 Constant Pressure Equilibria 2.2.2 Constant Temperature Equilibria 2.3 Relative Volatility and Its Derivation 2.4 Raoult's Law and Henry's Law 2.5 Maximum Boiling Azeotropes 2.6 Minimum Boiling Azeotropes 2.7 Methods of Distillation 2.7.1 Flash Vaporization 2.7.1.1 Material Balance and Calculation 2.7.2 Differential Distillation 2.7.2.1 Derivation of Rayleigh's Equation and Calculation 2.7.3 Continuous Rectification of Binary Solution 2.8 McCabe-Thiele Method 2.8.1 Derivation of Enriching and Stripping Section 2.8.2. Stepwise Procedure for obtaining Theoretical Trays by McCabe-Thiele Method 2.8.2 Calculation of Number of Trays	11	23

	2.9 Introduction of Feed and Location of the feed 2.10 Reflux Ratio 2.10.1 Total Reflux Ratio 2.10.2 Minimum Reflux Ratio 2.10.3 Optimum Reflux Ratio 2.11 Other Distillation Techniques 2.11.1 Steam Distillation 2.11.2 Vacuum Distillation 2.11.3 Azeotropic Distillation 2.11.4 Extractive Distillation 2.12 Reboilers and Their Use		
Unit – III Humidification	3.1 Vapor-Pressure Curve 3.2 Saturated and Unsaturated Vapor Gas Mixtures 3.3 Concept of Absolute Humidity, Relative Saturation, Percentage Saturation, Dew Point, Dry Bulb Temperature, Wet Bulb Temperature, Adiabatic Saturation Temperature, Humid Volume, Humid Heat, Enthalpy 3.4 Calculations of Absolute Humidity, Relative Saturation and Percentage Saturation 3.5 Psychometric Chart for Air-Water system 3.6 Purposes of Contact of Gas with Pure Liquid 3.7 Equipments 3.7.1 Cooling Towers – Natural and Mechanical Draft 3.7.2 Spray Chamber 3.8 Range, Approach and Cycle of Concentration for Cooling Tower	08	18
Unit – IV Drying	4.1 Drying Equilibrium 4.2 Definitions and Calculation of Moisture Content (Wet and Dry Basis), Equilibrium and Free Moisture, Bound and Unbound moisture 4.3 Classification of Dryers 4.4 Construction and Working of Drying Equipment 4.7.1 Tray Drier 4.7.2 Vacuum Drier 4.7.3 Rotary Drier 4.7.4 Spray Drier 4.5 The Rate of Batch Drying – Derivation and Calculations 4.8.1 Constant Rate Period 4.8.2 Falling Rate Period	08	18
Unit – V	5.1 Definition and Industrial Application of Adsorption 5.2 Types of Adsorption 5.3 Nature of Adsorbents and Commonly Used Adsorbents	08	18

Adsorption and Ion Exchange	5.4 Adsorption Equilibria 5.5 Adsorption Hysteresis 5.6 Effect of Temperature on Adsorption 5.7 The Freundlich's Adsorption Isotherm 5.8.1 Freundlich's Equation for Single Stage Operation 5.8.2 Material Balance 5.8 Equipment 5.9.1 Higgins Contactor 5.9.2 Pressure Swing Adsorber 5.10 Ion-Exchange: Principles and Its Application		
Unit – VI Crystallization	6.1 Crystallization and Industrial Applications 6.2 Equilibria and Yield of Crystallization 6.3 Methods of Super Saturation, 6.4 Nucleation and Crystal Growth 6.5 Crystallization Equipment 6.5.1 Vacuum Crystallizer 6.5.1 Swenson Walker Crystallizer. 6.6 Meir's Theory 6.7 Crystallization With and Without Seeding 6.8 Caking Of Crystals and Its Prevention 6.9 Calculation of Crystal Yield	06	14
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
25	**45**	**30**	---------	————————	————————

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S.No.	Title of Book	Author	Publication with place, year and ISBN
1	Unit Operations of Chemical Engineering	McCabe, Warren L., Julian C. Smith	McGraw Hill Publication, New York 2004 7th Edition
2	Mass Transfer Operations	Robert E. Treybal	McGraw- Hill, 3rd Edition, 1981
3	Separation Process Principles	Ernest J. Henley, J. D.Seader, D. Keith Roper	Wiley India, 2nd Edition, 2005

4	Unit Operation-II	K.A. Gavhane	NiraliPrakashan, Pune 2009
5	Unit Operations of ChemicalEngineering, Volume-1	P. Chattopadhyay	Khanna Publishers, New Delhi, 1995
6	Introduction to Chemical Engineering	L.Badger, Julius T. Banchero	McGraw Hill Publication, New York, 7th Edition, 2004

(b) Open-source software and website:

1. https://ndl.iitkgp.ac.in/
2. https://www.vlab.co.in/
3. https://swayam.gov.in/
4. https://onlinecourses.nptel.ac.in
5. www.unitoperation.com
6. MIT Open Learning Library

Suggested Practical List:

Sr. No.	Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology)	Approx. Hrs Required
1	Demonstrate principle, construction and working of equipments for gas–liquid operations with models	2
2	Prepare vapour liquid equilibria curve at atmospheric pressure for Benzene-Xylene	2
3	Carry out simple distillation in glass assembly	2
4	Find out the effect of vacuum on distillation of liquid	2
5	Carry out continuous rectification in packed column	2
6	Find out amount of steam required in steam distillation	2
7	Find out the property of atmospheric air with the help of wet bulb and dry bulb temperature	2
8	To determine the range, approach and efficiency of the cooling tower.	2
9	Prepare drying curve of moist sand and moist limestone	2
10	Find out equilibrium moisture content and drying time of wet solid	2
11	To determine the drying characteristic for rotary dryer	2
12	Characterize industrial adsorbents and observe their samples	2
13	Remove colour impurities from water using charcoal	2
14	Find out the yield of crystals from saturated solution without seeding	2
15	Find out the yield of crystals of from saturated solution with seeding	2

Suggested Activities for Students:

1. Assignments
2. Technical Quiz/MCQ Test
3. Preparation of non-working models of various mass transfer equipments and its importance.
4. Preparation of power-point slides, which include videos, animations of various mass transfer equipments.

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